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PUBLIC ECONOMICS

Assignment # 2

A1: Two goods: good 1 and good 2.
- One consumer
- One firm

$U = c_1 c_2$, for consumer

$X_2 = \frac{X_1}{c}$, for producer

Given: $P_1 = 1$

So, let $P_2 = p$.

a) We know,

At the equilibrium,

$$MRS = MRT$$

$$\therefore \frac{P_x}{P_y} = \frac{dX_2}{dX_1}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{dX_2}{dX_1}$$

$$\Rightarrow \frac{1}{p} = \frac{1}{c}$$

$$\therefore p = c$$

Hence, the equilibrium price of good 2 is c, where c is the given constant, in the transformation f^n.

Budget constraint: $P_1 c_1 + P_2 c_2 \leq w$

$\Rightarrow c_1 + c c_2 \leq 1$ (Initial endowment)

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b) We need to maximise

$$u = c_1 c_2$$

$$\text{s.t. } c_1 + c c_2 \leq 1$$

At point of maxima, condition changes to:

$$c_1 + c c_2 = 1$$

Since it is a constrained optimisation problem, we can use Lagrange Multipliers.

$$L = c_1 c_2 + \lambda [1 - c_1 - c c_2]$$

diff w.r.t c_1, c_2 and λ,

$$\frac{dL}{dc_1} = c_2 + \lambda(-1) \leq 0$$

$$\frac{dL}{dc_2} = c_1 + \lambda(-c) \leq 0$$

$$\frac{dL}{d\lambda} = 1 - c_1 - c c_2 \leq 0$$

However, at the optimal point, the equality must hold So,

$$c_2 = \lambda \quad \text{—— (I)}$$

$$c_1 = c\lambda \quad \text{—— (II)}$$

$$c_1 + c c_2 = 1 \quad \text{—— (III)}$$

Sub (I) and (II) in (III)

$$c\lambda + c\lambda = 1$$

$$\Rightarrow 2c\lambda = 1$$

$$\Rightarrow \boxed{\lambda = \frac{1}{2c}}$$

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∴ From (1),

$$c_2 = \frac{1}{2c}$$

From (II)

$$c_1 = c\left(\frac{1}{2c}\right) = \frac{1}{2}$$

Hence,

$$U = \left(\frac{1}{2c}\right)\left(\frac{1}{2}\right)$$

$$\Rightarrow U = \frac{1}{4c}$$

c) We know that for an indirect tax like this one,

$$\frac{dP}{dt} = \frac{\epsilon_D}{\epsilon_S - \epsilon_D}$$

This means that the change in price after tax is simply a function of the elasticities of the producer and consumer and does not depend on who is liable to pay the tax.

In other words, legal incidence is independent of economic incidence

An interesting observation here is that

$$\text{Tax Incidence} \propto \frac{1}{\text{Elasticity}}$$

for both parties

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d) We see that if the firm receives even ϵ less than c for good 2, it will stop the transformation of X_1 to X_2.
So, we can say that its elasticity is ∞.
The entire tax burden is on the consumer.

Budget constraint $c_1 + (c+t)c_2 = 1 + T$

X Note that this isn't the correct budget constraint as T is not available to the consumer initially.

So,
Actual budget constraint:

$$c_1 + (c+t)c_2 = 1$$

We need to maximise

$$c_1 c_2$$

$$\text{s.t. } c_1 + (c+t)c_2 = 1$$

e) Using Lagrange

$$\mathcal{L} = c_1 c_2 + \lambda[1 - c_1 - (c+t)c_2]$$

$$\frac{dL}{dc_1} = c_2 + \lambda(-1) = 0$$

$$\frac{dL}{dc_2} = c_1 + \lambda[-(c+t)] = 0$$

$$\frac{dL}{d\lambda} = 1 - c_1 - (c+t)c_2 = 0$$

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$\therefore \quad c_2 = \lambda \quad \text{—— (I)}$

$c_1 = \lambda(c+t) \quad \text{—— (II)}$

$c_1 + (c+t)c_2 = 1 \quad \text{—— (III)}$

Sub (I) and (II) in (III)

We get,

$$\lambda(c+t) + (c+t)\lambda = 1$$

$$\Rightarrow 2\lambda(c+t) = 1$$

$$\Rightarrow \lambda = \frac{1}{2(c+t)}$$

Hence,

$$c_1 = \frac{c+t}{2(c+t)} = \frac{1}{2}$$

$$c_2 = \lambda = \frac{1}{2(c+t)}.$$

Thus is <u>not</u> pareto optimal

$$c_2 = \frac{1}{2(c+t)} < \frac{1}{2c}$$

So consumer utility goes down and producer revenue also goes down.

The T money has not been fully utilised and hence we have been unable to reach an optimal solution.

Clearly the answer in (b) makes both parties better off.

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f) Now we notice that the consumer has received a transfer of $T = c_2 t$

So, he can use this as a fresh endowment and using Lagrange, we find:

$$c_1 = \frac{c_2 t}{2}$$

$$c_2' = \frac{c_2 t}{2(c+t)}$$

This forms a never-ending story of buying, transfer and more buying.
So we can find a pattern here:

$$c_{1net} = \frac{1}{2} + \frac{c_2 t}{2} + \dots$$

$$c_{2net} = \frac{1}{2(c+t)} + \frac{t}{2(c+t)}\left(\frac{1}{2(c+t)}\right) + \frac{t}{2(c+t)}\left(\frac{t}{2(c+t)}\right)\left(\frac{1}{2(c+t)}\right)$$

So $= \frac{a}{1-r} = \frac{1/2(c+t)}{1 - t/2(c+t)}$

$$= \frac{1/\cancel{2(c+t)}}{2(c+t) - t/\cancel{2(c+t)}}$$

$$c_{2net} = \frac{1}{2c+2t-t} = \underline{\underline{\frac{1}{2c+t}}}$$

We know,

$$\frac{c_{1net}}{c_{2net}} = c+t$$

$$\Rightarrow c_{1net} = \frac{c+t}{2c+t}$$

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$\therefore$ Max utility $= C_{net} \cdot S_{net}$

$$= \frac{c+t}{(2c+t)^2}$$

Welfare loss $= \frac{1}{4c} - \frac{c+t}{(2c+t)^2}$

$$= \frac{4c^2 + t^2 + 4ct - c - t}{(4c)(2c+t)^2}$$

★ Since the transfer is happening to the consumer itself, who is also facing the entire tax burden, the best solution would be to just remove the tax altogether.

A2: $u(c, l) = \frac{c^{1-\theta}}{1-\theta} - l$

c: consumption
l: labour supply
w: wage rate
t: tax rate

a) BUDGET CONSTRAINT

$$c = lw(1-t)$$

Optimal labour supply f^n:

$$\max_{c, l} \frac{c^{1-\theta}}{1-\theta} - l$$

$$s.t. \quad lw(1-t) = c$$

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So we need to maximise

$$\frac{[l w (1-t)]^{1-\theta}}{1-\theta} - l$$

Diff. w.r.t l
we get,

$$\cancel{(1-\theta)}\frac{[l w (1-t)]^{-\theta}}{\cancel{1-\theta}} w(1-t) - 1$$

FOC condition:

$$[l w (1-t)]^{-\theta} w(1-t) = 1$$

$$\Rightarrow [l w (1-t)]^{-\theta} = \frac{1}{w(1-t)}$$

$$\Rightarrow [l w (1-t)]^{\theta} = w(1-t)$$

$$\Rightarrow l w (1-t) = [w(1-t)]^{1/\theta}$$

$$\therefore l = [w(1-t)]^{1/\theta - 1}$$

b) $\log l = \frac{1-\theta}{\theta}\left(\log(1-t) + \log w\right)$

$$\therefore \frac{1}{l}\frac{dl}{d(1-t)} = \frac{1-\theta}{\theta}\left[\frac{1}{1-t}\right]$$

$$\therefore \frac{dl/l}{d(1-t)/1-t} = \frac{1-\theta}{\theta}$$

Hence, Proved.

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c) Since the economies are similar, we can say that the number of people involved in the labour market are roughly the same

Hence, average tax rate (new) $= \frac{35 + 28.5}{2}$

$= 31.75\%$

So, $\frac{d(1-t)}{1-t} = \frac{(1-t') - (1-t)}{1-t}$

$= \frac{t - t'}{1-t}$

$= \frac{(35 - 31.75)/100}{1 - 0.35}$

$= \frac{0.0325}{0.65} = \frac{1}{20}$

We saw

$\frac{d(1-t)}{1-t} \cdot \frac{l}{dl} = \frac{\theta}{1-\theta}$

$\therefore$ ~~$\frac{\theta}{\theta} = \frac{1}{20}$~~

$\frac{dl}{l} = \frac{4400 - 4000}{4000} = \frac{400}{4000} = 0.1$

$\therefore \frac{d(1-t)/d(1-t)}{dl/l} = \frac{1/20}{1/10} = \frac{1}{2}$

$\Rightarrow \frac{1-\theta}{\theta} = 2$

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d) Obviously, the revenue maximising tax rate has to be less than 100%.
Otherwise, it would prevent people from working as they'd get no money out of it.
Hence the government would get zero revenue.
So, this is clearly a BAD CHOICE.

Revenue = lwt

$$R = tw\left[w(1-t)\right]^{\frac{1-\theta}{\theta}}$$

$$\frac{dR}{dt} = w\left[w(1-t)\right]^{\frac{1-\theta}{\theta}} + tw\left(\frac{1-\theta}{\theta}\right)\left[w(1-t)\right]^{\frac{1-\theta}{\theta}-1}(-w)$$

$$= w\left[w(1-t)\right]^{2} + 2tw\, w(1-t)\cdot(-w) = 0$$

$$\Rightarrow \cancel{w^3}(1-t)^{\cancel{2}} = 2t\cancel{w^3}\cancel{(1-t)}$$

$$1-t = 2t$$

$$3t = 1$$

$$\Rightarrow t = \frac{1}{3} \text{ or } 33.3\%$$

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